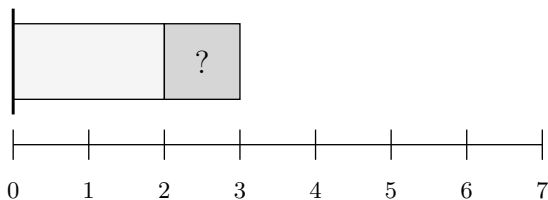
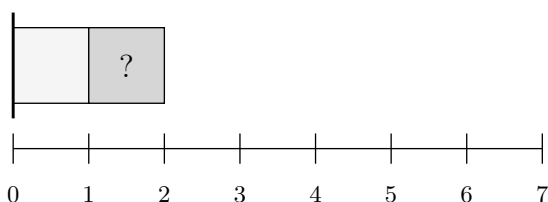


A hidden end can be anywhere in its gray band, including the edges. Each trial may use new lengths. Join strips end to end from 0, with no gap or overlap.

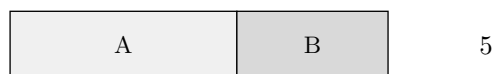
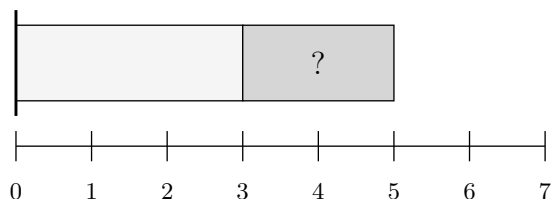
A: 2 to 3 units



B: 1 to 2 units

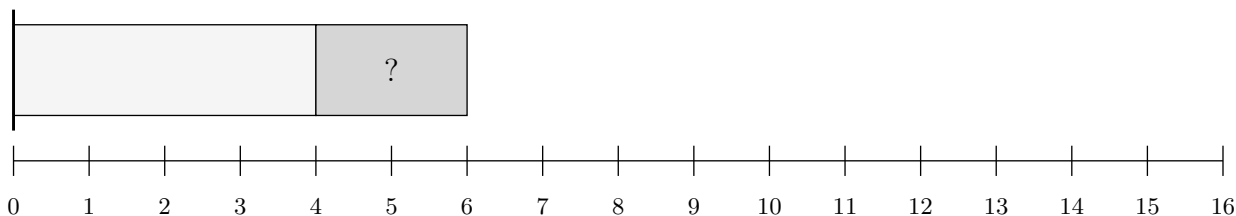


A + B: 3 to 5 units

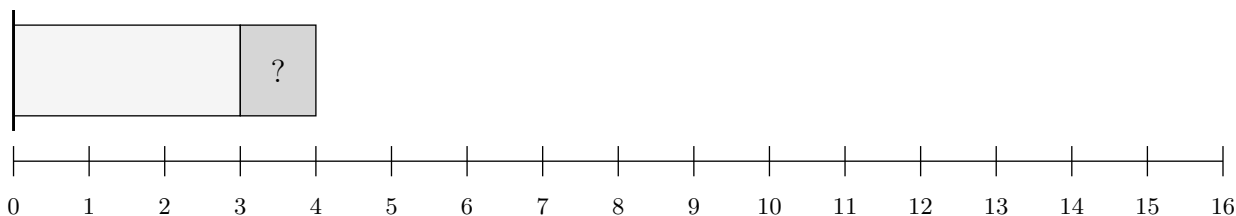


Problem 1: Make the shortest and longest joins with these strips. Can you make a join ending exactly at 8?

A: 4 to 6 units

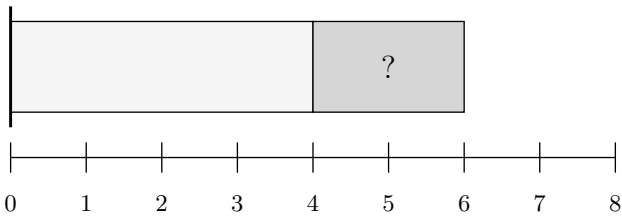


B: 3 to 4 units

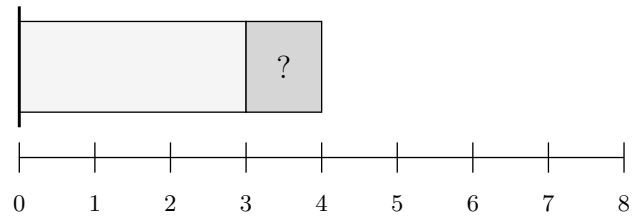


Problem 2: Try A with B, C with D, and E with F; which pairs always reach 8?
Let a friend try to defeat your choices.

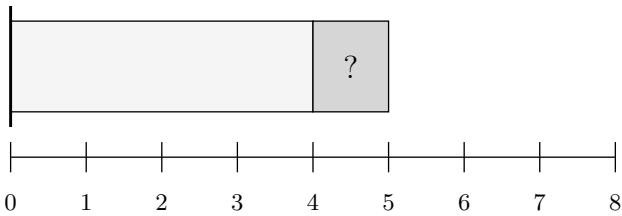
A: 4 to 6 units



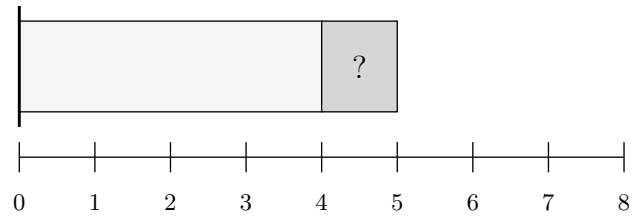
B: 3 to 4 units



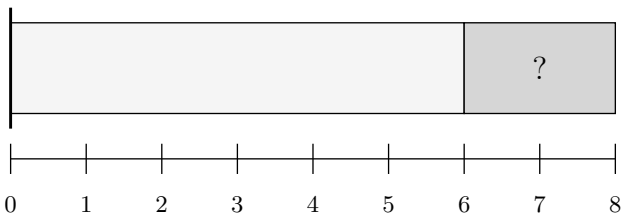
C: 4 to 5 units



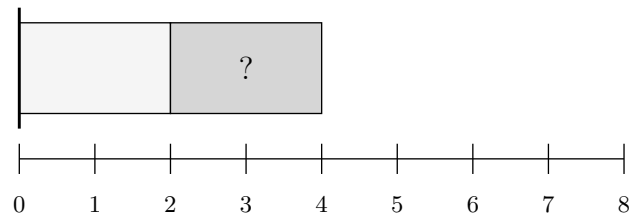
D: 4 to 5 units



E: 6 to 8 units



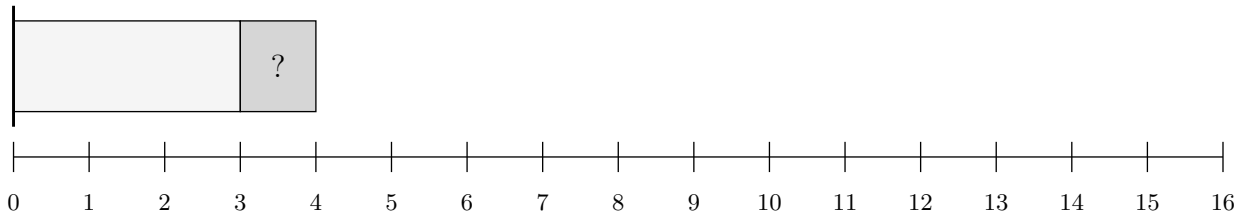
F: 2 to 4 units



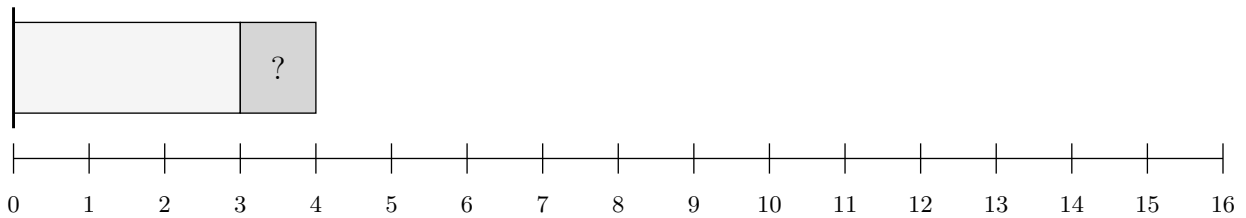
Reaching 8 includes ending at 8.

Problem 3: Pick a card for B so the join always reaches 7 and fits within 9. Try to break your choice with different allowed strips.

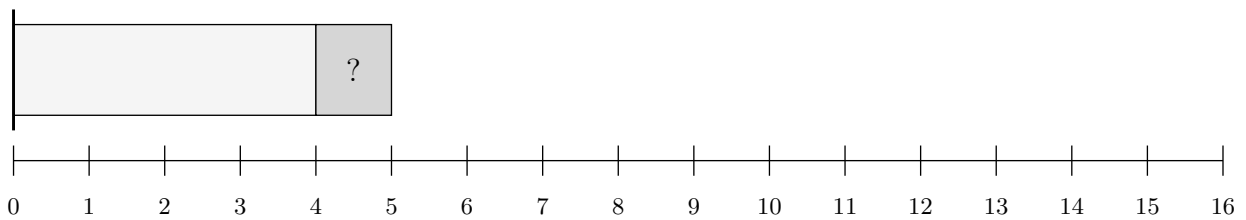
A: 3 to 4 units



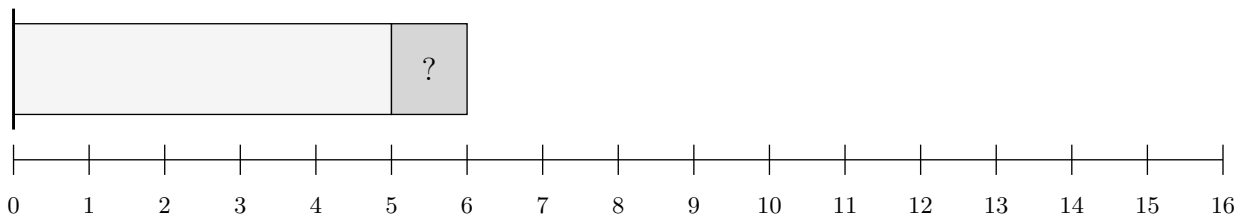
B: 3 to 4 units



B: 4 to 5 units

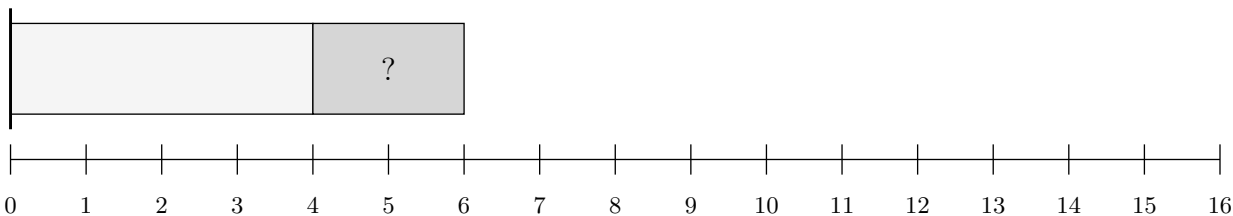


B: 5 to 6 units

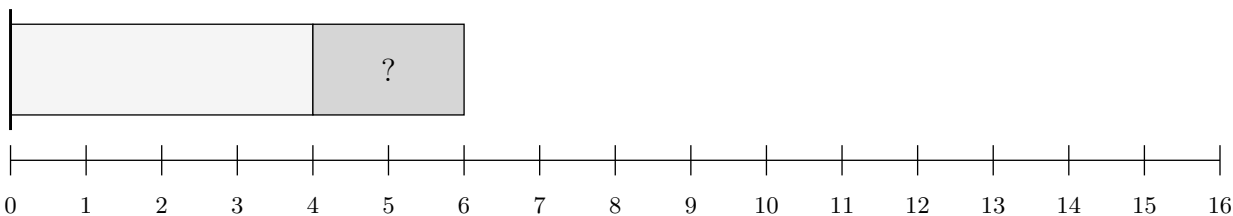


Problem 4: Find how small and how big each end gap can be. Start at 0 and keep A unchanged within each comparison; B may vary separately.

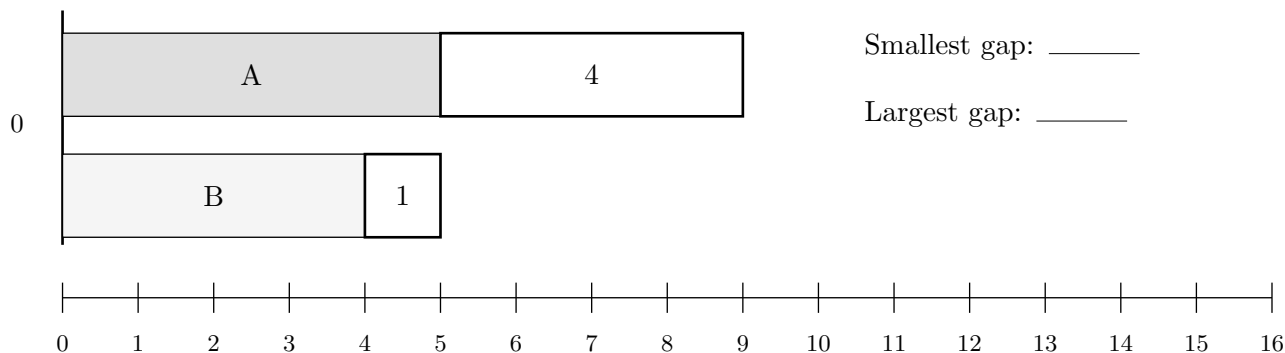
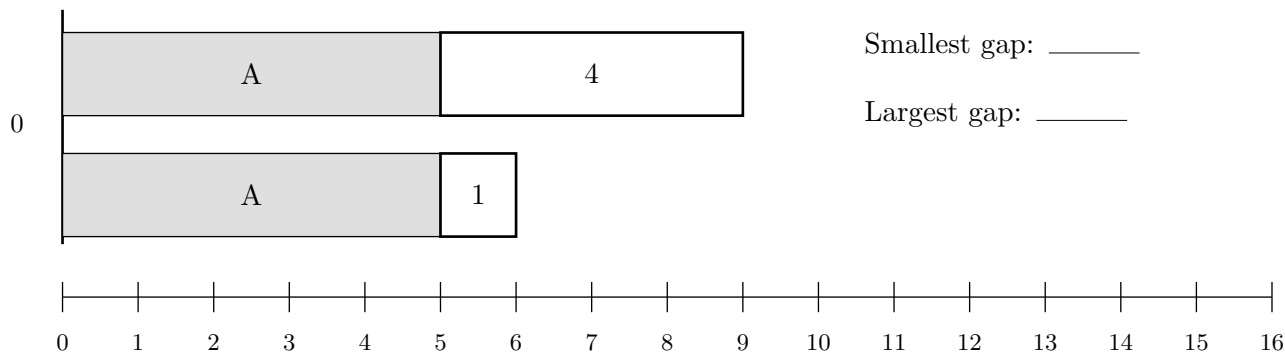
A: 4 to 6 units



B: 4 to 6 units

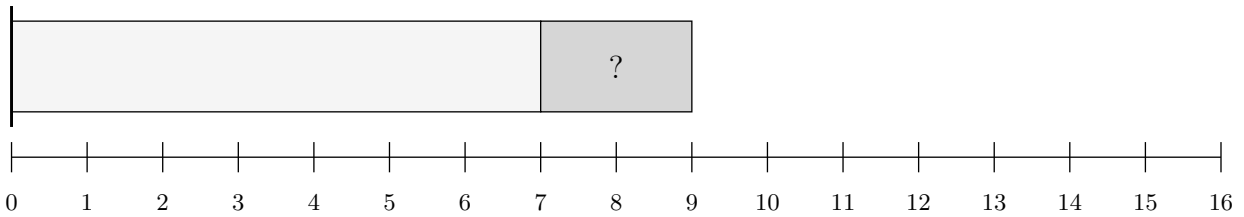


One possible setting; the white pieces are exactly 4 and 1 units long.

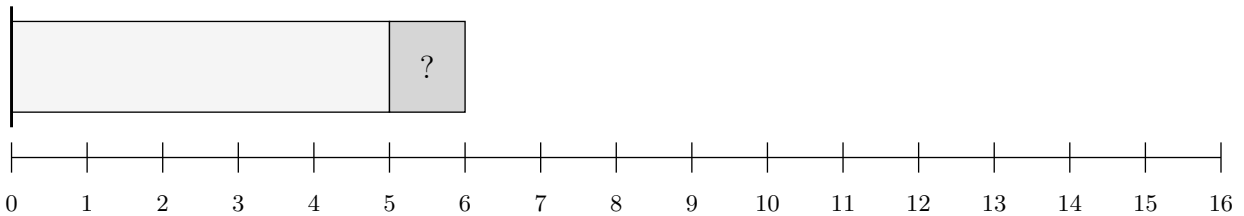


Problem 5: Put both starts at 0. Make the biggest gap between the ends, then the smallest.

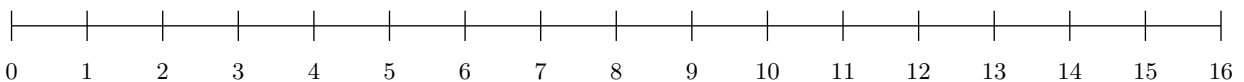
long strip: 7 to 9 units



short strip: 5 to 6 units

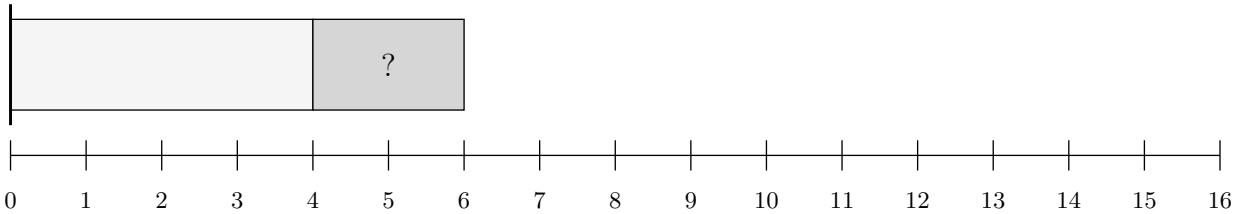


Problem 6: Using these two strips, make a gap of 3 in two different ways.



Problem 7: A and B must fill 9 units exactly. Find every pair of whole-unit lengths that fits both cards.

A: 4 to 6 units



B: 3 to 5 units

